



challenge 1





Challenge: Capture the constraints over the infinite space



Priority Work/Questions: MONDRIANAL'S

Finite space: $(2^q)!$ permutations



infinite spaces:



x



f



x



$\varphi(x), v,$
 $\theta), f(x), v,$

complex amplitudes: $z_i / \varphi_i = 1$











$$|0\rangle \rightarrow (|0\rangle + |1\rangle\sqrt{2})$$

$$11 \rightarrow 10 - 11112$$





$$|0\rangle \rightarrow \cos \theta |0\rangle + \sin \theta |1\rangle$$

$$|1\rangle \rightarrow -\sin\theta |0\rangle + \cos\theta |1\rangle$$



••• Infinite Interpretations











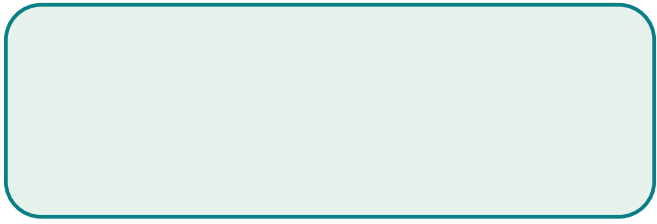






2 finite interpretations for 1 qubit









G

1



G

2





G

n







G

1



G

2





G

n











Q1















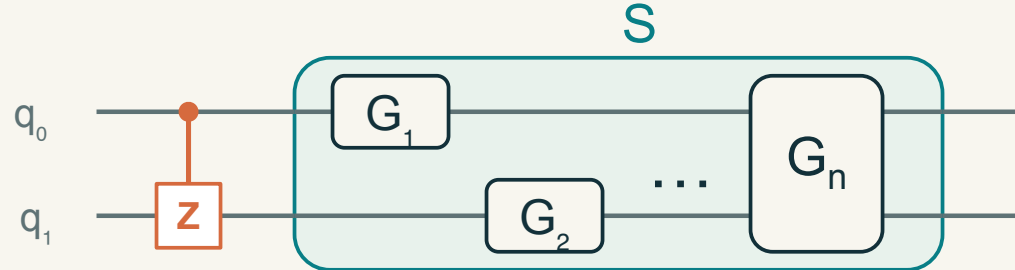








Challenge 1



Prior Work(Queso): MONOMIAL S

Finite space: $(2^n)!$ permutations

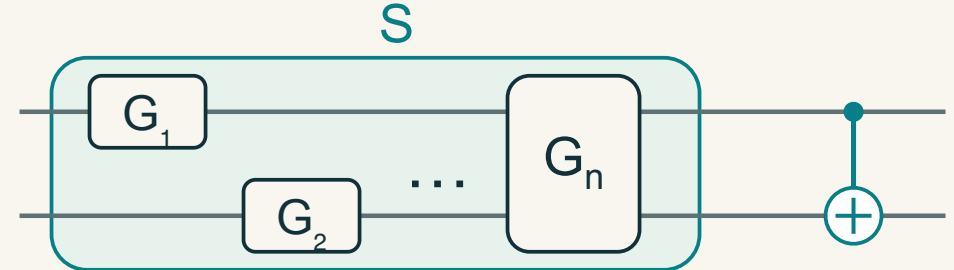
$$|x\rangle \rightarrow |f(x)\rangle$$

$$\left\{ \begin{array}{l} |0\rangle \rightarrow |0\rangle \\ |1\rangle \rightarrow |1\rangle \end{array} \right\}$$

$$\left\{ \begin{array}{l} |0\rangle \rightarrow |1\rangle \\ |1\rangle \rightarrow |0\rangle \end{array} \right\}$$

2 finite interpretations for 1 qubit

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GENERAL S · SUPERPOSITION ALLOWED

Infinite space:

$$|x\rangle \rightarrow \sum_{y \in \mathbb{Z}_2} \varphi(x, y, \theta) |f(x, y)\rangle$$

Complex amplitudes: $\sum_i |\varphi_i|^2 = 1$

$$\left\{ \begin{array}{l} |0\rangle \rightarrow (|0\rangle + |1\rangle)/\sqrt{2} \\ |1\rangle \rightarrow (|0\rangle - |1\rangle)/\sqrt{2} \end{array} \right\}$$

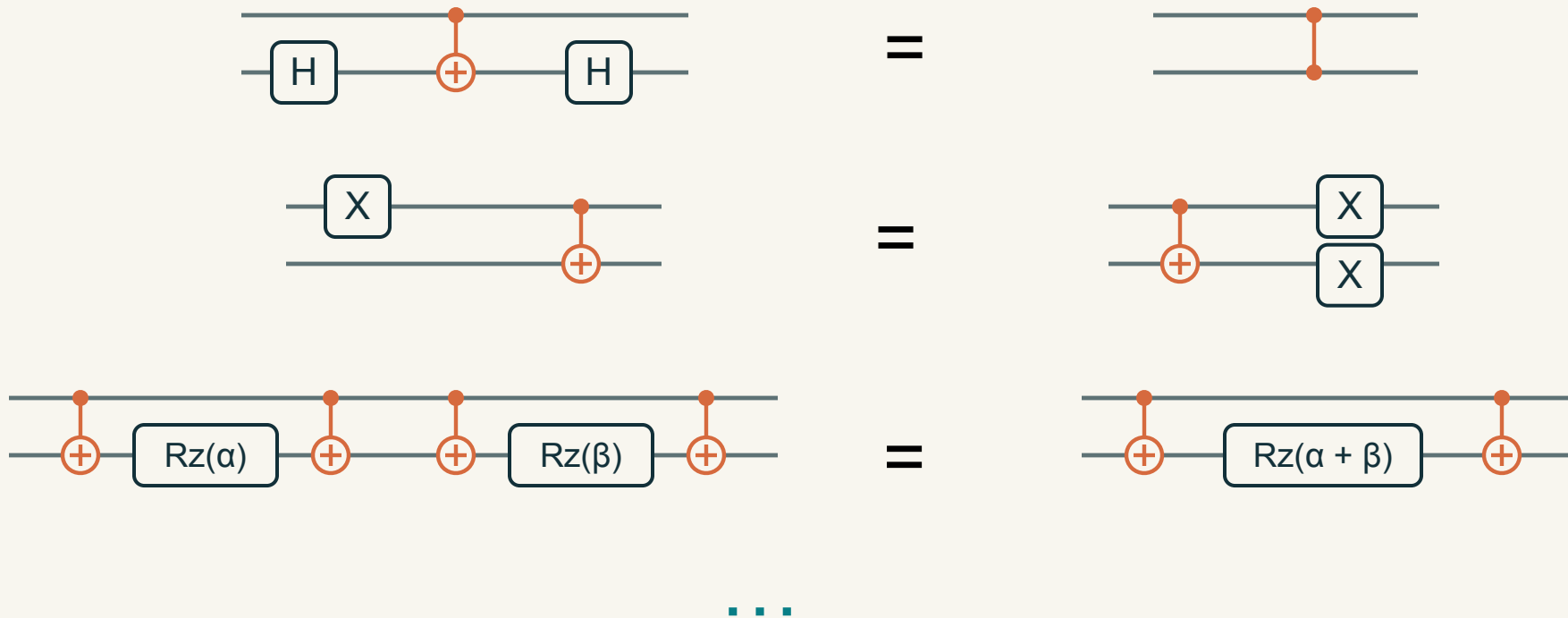
$$\left\{ \begin{array}{l} |0\rangle \rightarrow \cos \theta |0\rangle + \sin \theta |1\rangle \\ |1\rangle \rightarrow -\sin \theta |0\rangle + \cos \theta |1\rangle \end{array} \right\}$$

... Infinite Interpretations

Challenge: Capture the constraints over the infinite space of S .

Challenge 2: Large Search Space

Many(infinite) equivalences even for concrete rewrite



Challenge: Hard to provide guaranteed and quantified coverage